

Supplementary materials for
Profiling the Chinese Participants in Psychological Science: Non-WEIRD but
Not Representative

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S1. Supplementary methods: Details of 1000 Chinese papers

The first data source of our study is derived from five mainstream Chinese journals. Of 1000 empirical papers, 500 were published between 2017 and 2018 that were selected by Wang et al. (2021). Another 500 papers were selected from the same five journals but published at different time points. The criteria and procedure of article selection are consistent with those of Wang et al. (2021) and are detailed as follows:

Step 1: Assigning identifiers to articles. For papers published in different periods, we obtained information of all the papers published in those journals across three different periods and assigned a unique identifier to each article. Each article ID has 8 digits. The first four numbers represent the selected period (we used 2008, 2018, 2021 to represent articles from three different periods), the fifth number represents the journal ID, from 1 to 5, and the last three numbers represent the order of the paper in the journal. For example, the first article in *Acta Psychologica Sinica* from 2008 is coded as 20081001, the second article in *Acta Psychologica Sinica* in the same year is coded as 20081002. All articles and their IDs can be found at <https://osf.io/avb7t/>.

Step 2: Random sampling from all articles. We used the `'sample'` function of R base to randomly select a certain number of papers from all the papers in each journal (see the code at <https://osf.io/avb7t/>). The number of papers sampled from each journal was weighted by the total number of papers published in that year (the total papers each period and the number of sampled papers, see Table S1). After getting the identifiers of the selected papers, two independent researchers checked each article to make sure that it was an empirical study. If not, we replaced the article with the empirical article, which had the smallest distance to the article whose identifier was sampled.

Table S1. The number of articles in five Chinese psychological journals

Journal	2008	2017~2018	2020~2021
<i>Acta Psychological Sinica</i>	138 (39)	246 (97)	91 (28)
<i>Journal of Psychological Science</i>	379 (107)	299 (115)	203 (61)
<i>Chinese Journal of Clinical Psychology</i>	227 (64)	379 (146)	310 (94)
<i>Psychological Development and Education</i>	87 (24)	162 (61)	95 (29)
<i>Psychological and Behavioral Studies</i>	57 (16)	213 (81)	125 (38)

Note: Each column includes the total number of articles published in each journal at that time interval and the number of articles selected (inside parentheses)

S2. Supplementary methods: The simulation of Stage1 for sex ratio

We conducted a simulation to check whether the non-informative prior is reasonable when using the criterion of $BF \geq 6$. Firstly, we chose the smallest effect size of interest based on data from Rad et al. (2018). From the supplementary material of Rad et al. (2018), we found 21 papers (33 studies in total) that reported sex ratios. From these data, we calculated the mean and 95% CIs of the absolute deviation of sex ratio from a hypothesized expected sex ratio of 0.5, which revealed that the mean of absolute deviation was 0.085, 95% CIs were [0.061, 0.115]. We used the lower limit of the 95% CIs as the smallest effect size of interest. Then, we conducted the simulation using two different strategies. One approach fixed prior to non-informative and varied the sample size N , and the other fixed the sample size N and varied the prior. Our results showed that for the fixed prior simulation, when sample size $N \geq 1200$, the proportion of getting evidence for both H_0 and H_a was greater than 80%. However, for the simulation with fixed N , varying prior did not increase the proportion of getting evidence for H_a .

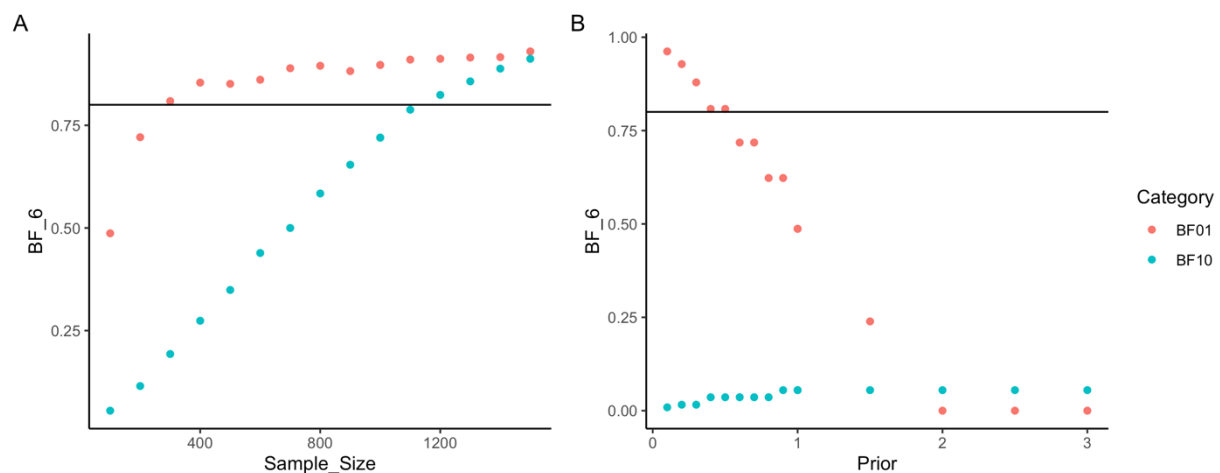


Fig S1. Simulation results. (A) Simulation with fixed prior and varying N , the x-axis is sample size N , and the y-axis is the proportion of BF values that are greater than 6. (B) Simulation with fixed N , the x-axis is the prior, the y-axis is the proportion of BF values that are greater than 6. Note: BF₀₁ in the figure corresponds to BF_{0a}, and BF₁₀ corresponds to BF_{a0}.

S3. Supplementary methods: The simulation of Stage1 for age distribution

To test the minimal effect that the Bayesian multinomial test can detect with non-informative prior and sample size $N \geq 1200$, we conducted a simulation with a simple assumption that age data were normal distribution for both the observed and the expected data. The simulations were as follows:

Step 1: We estimated the mean (23.01) and *SD* (4.65) of Chinese participants' age from Jones et al. (2021). These data provided a starting point for simulation.

Step 2: We used the mean age 23.01 as the mean age of observed, and we chose a mean age difference (e.g., *diff_age* = 1 year), the mean age of the second group was 23.01 + *diff_age*. We assumed that both groups shared the same *SD*.

Step 3: We generated multinomial data in the same way we generated count data for different age bins when individual data is not available. For example, we generated the count number of participants under 18 by using `round((pnorm(18, mean = Mean1, sd = SD1) * 1200))`, and the counted number of participants between 18 and 26 by using `round((pnorm(26, mean = Mean1, sd = SD1) * 1200)) - round((pnorm(18, mean = Mean1, sd = SD1) * 1200))`. Replacing "Mean1" and "SD1" we obtained the data for group 1 and group 2. If the count data was zero for certain age bin, we replaced zero with one to avoid error when calculating the Bayes factor.

Step 4: We conducted the Bayesian multinomial test, using data from group 1 as the observed and data from group 2 as the expected. We saved the BF_{a0} and $\ln(BF_{a0})$ resulting from this step.

Step 5: We repeated the Step 2 to Step 4, iterating the *diff_age* from 0 to 2, with the interval of 0.05, and saved the BF_{a0} and $\ln(BF_{a0})$ for each mean age difference.

Finally, we repeated the above Step 2 to Step 5 with different mean age of group 1 from 10 to 60, with 5 years as the interval, and fixed the *SD* as 4.65, we found that when the mean age was moving toward two ends (0 or 60), the current setting was no longer able to detect the mean age difference. However, this pattern is understandable because when the mean age is moved closer to either end of the ages, most of the generated count data fell in the age bin at that end and few data in other age bins.

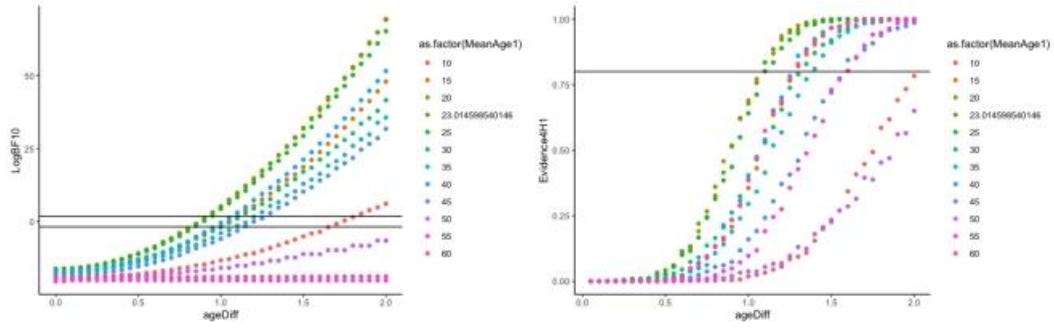


Fig S2. (A) Simulated results of $\ln(BF_{a0})$ (y-axis) for different mean age differences (x-axis) and different mean ages (colors of dots). The two black horizontal lines represent $\ln(BF_{a0}) > \ln(6)$ (upper) and $\ln(BF_{a0}) < \ln(1/6)$ (lower). (B) Results of the alternative simulation with sample size $N = 1200$. The x-axis is the differences between the means of group 2's age and group 1's age, different colors represent different means ages from group 1. The y-axis is the percentage of $BF \geq 6$ in 1000 simulations. Note: LogBF_{10} in the figure corresponds to $\ln BF_{a0}$, and "Evidence4H1" corresponds to "Evidence for H_a ".

The above simulation suggested that our Bayesian Multinomial test with non-informative prior and $N = 1200$ can detect a mean age difference of 0.95 to 1.35, when the mean age was between 15 years old and 45 years old (See Fig S2 A). Please note that a mean age of 15 to 45 is consistent the observation from the literature. For instance, in the supplementary of Rad et al. (2018), 21 out of 33 studies investigated young and mid-aged adults, with mean age range from 18.63 to 36.81. The mean age difference of 0.95, to be expressed more intuitively, is corresponding to Cohen's $d = 0.2041$, and the mean age 1.35 corresponds to Cohen's $d = 0.29$, if we assumed the age data are normally distributed with $SD = 4.65$. Also, note that the simulation above lacked stochastic process, each mean age difference only returned one BF value, we conducted an alternative simulation, in which we generated age data and converted these age data into age bins, instead of generating count data for each age bin directly, the results revealed that, when the mean age was from 15 to 45, the current setting can detect a mean age difference from 1.1 years to 1.6 years, corresponding to Cohen's d of 0.236 to 0.344 (Figure S2 B, see our online Rmarkdown for details:

https://github.com/Chuan-Peng-Lab/chin-subj/blob/master/1_Protocol/1_2_Reg_Report_Stage_1/1_2_4_Reg_Report_Stage_1_Analysis/Notebook_Data_Analysis_CHN_Sample_Stage1_RR.nb.html).

S4 Supplementary results: Deviation from the Stage 1 protocol

Table S2. *Preregistration Deviations Table Template*

Deviations					
#	Details		Original Wording	Deviation Description	Reader Impact
1	Type	Data Preparation	The original wording can be found <i>S1</i> , “Step 2”.	The 2018 data were subject to coding discrepancies: <i>Acta Psychological Sinica</i> had 2 overcoded articles; <i>Psychological Development and Education and Psychological</i> and <i>Psychological and Behavioral Studies</i> each had 1 undercoded article.	This deviation has no impact on readers’ interpretations of the findings, as the coding bias occurred in only two articles and the coding data were generated prior to data analysis.
	Reason	Miscommunication			
	Timing	During data collection			
2	Type	Data Preparation	<i>These projects were chosen because they opened raw data.</i>	For some international collaborative projects, the raw data either lacks clear acquisition procedures or has ambiguous definitions. We attempted	This deviation has a small impact on readers’ interpretations of the findings. This is because the number of datasets included in our analysis is more
	Reason	New knowledge			

	Timing	During data collection		to access these data via multiple rounds of email requests. However, we were unable to obtain the demographic data or the clear variable definitions from several projects. Therefore, these projects were excluded from our analysis.	than those that are publicly available, and the overall sample size is large enough for testing our hypothesis.
3	Type	Variables	<i>We will divide all the participants into four categories: stated specific population; inferred specific population; inferred general population; stated general population.</i>	We used three categories: stated specific population, inferred general population, and stated general population.	This deviation has a small impact on readers' interpretations of the findings. We deleted the category "inferred specific population" because, during the coding, we found that studies that implied target at an "inferred specific population" do not exist, and therefore this category is redundant.
	Reason	Other (Please Explain)			
	Timing	Before data collection			
4	Type	Data Preparation	<i>To ensure the accuracy of the coding content, the results from the two coders are compared after completing the initial coding. Two coders rate the consistency of each article from 0 to 1, with 0 represents completely different and 1 represents</i>	We used the R package <i>irr</i> to calculate the coding reliability score. Instead of relying on coders' ratings for the calculation, we computed it directly by text-matching based on the coding content. For specific	This deviation does not affect readers' interpretation. This is because the consistency between coders was rated automatically, which decreases human error and aligns with common practices in current research (Hayes &
	Reason	Other (Please Explain)			
	Timing	Before data			

		collection	<i>identical. This consistency score will be then used for calculating the inter-rater reliability. We will use the R package irr for this index (Gamer et al., 2019).</i>	details, please refer to Sections 3.1 in the main text.	Krippendorff, 2007; Nili et al., 2020; Wu et al., 2025).
5	Type	Data Preparation	This part of the description is mainly derived from the coding manual V4, and the text is as follows: <i>(1) To further increase the reliability and objectivity of the coding content, we will redistribute the articles for proofreading after the coding stage is completed.</i> <i>(2) The coders will be divided into groups, with two coders in each group proofreading the same documents. If there is an inconsistency between two independent coders, we will go back to the article and resolve the inconsistency through discussion. If an agreement cannot be achieved between two coders, a third coder, a senior author, will join the discussion to make a final decision.</i>	The description in the main text is as follows: <i>During the coding proofreading stage, the reliability of the final coding results is ensured through a combination of automatic text-match (see script “Coding_check.R”) and manual check. Specifically, the consistency of each item was automatically rated as one of three levels by matching the corresponding text: (1) complete inconsistency, which means the text by two coders has no overlap, (2) partial consistency, which means the text by two coders are partially overlapped, and (3) complete consistency, which means the text by</i>	This change does not affect readers’ understanding. This change increases the efficiency of the coding and decreases human error, as we mentioned in the above deviation. Moreover, a manual check in the original plan was executed when inconsistencies were found.
	Reason	Plan not possible			
	Timing	During data collection			
	Reason	Other (Please Explain)			
	Timing	After data access			

				<i>two coders are exactly the same. This text-matching was then manually checked because the algorithm may not work well for Chinese characters. If an item was rated as inconsistent between coders (rated as 1 or 2), coders revisited the original text together to resolve discrepancies. Any remaining discrepancies, if present, were addressed via a final group consultation involving the project leader.</i>	
6	Type	Analysis	<i>Additionally, we will export the keywords of Chinese journals' articles and the big team projects through CNKI (China National Knowledge Infrastructure) and Web of Science, and then, if feasible, try to use bibliometric methods to compare the similarities and differences between the keywords of the big team projects and Chinese journals' articles.</i>	After coding, we found that keywords were available only from 14 (out of 27) large-scale international collaborative projects, and there was a substantial disparity in sample size between the two data sources; no comparison of similarities and differences was conducted. We described the abstract	This deviation has no impact on readers' interpretations of the research findings. The bibliometric analysis was exploratory and independent from our three main hypotheses.
	Reason	Other (Please Explain)			
	Timing	After data access			

				by using word cloud in the supplementary results (<i>Fig S13</i>)	
Unregistered Steps					
#	Details		Original Wording	Unregistered Step Description	Reader Impact
1	Type	Analysis	<i>we will use Bayesian multinomial test (corresponding to frequentists' goodness-of-fit test or χ^2) to test whether data from two sources differ from each other on certain dimension (e.g., sex, age, education attainment).</i>	As the education information was not available for many papers, we used dichotomous education levels to maximize the number of papers included in analysis. More specifically, we used two ways to categorize participants' educational attainment: less-than-college vs college-or-above, less-than-high-school vs high-school-or-above.	This method improves the robustness of our analysis. Because educational classification systems vary across countries and regions, and educational information is missing from some articles or projects, using these two different ways allows us to include more data in our comparison.
	Timing	After data access			
2	Type	Analysis	<i>For the age distribution, we created age bins using two different approaches. The first approach was based on the developmental stages...The second approach was similar to census data's age bins...Note that a more fine-grained age</i>	Stage 1 methods describe three age bins: five age bins, seven age bins and ten age bins. The first two were used for Bayes factor analysis, and the last one for plotting the population pyramid. In Stage 2, we	This deviation does not affect readers' interpretation of the results. In Stage 1, we pre-registered three age bins: five, seven and ten age bins, all of which were retained and reported in Stage 2. The newly added four age bins further
	Timing	After results known			

			<i>bins (with 5-year intervals) were only used for pyramid plot but not for statistical analysis.</i>	adopted all three above grouping methods and additionally introduced four age bins in H3 for clearer data presentation.	highlight data differences.
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S5. Supplementary results: The demographic reported in different dimensions

Demographic reporting has long received attention (Rad et al., 2018; Rozin, 2001), and there may be some changes in the reporting rate for Chinese psychological articles across different periods. Therefore, we visualized the demographic reported across different periods (Fig S3).

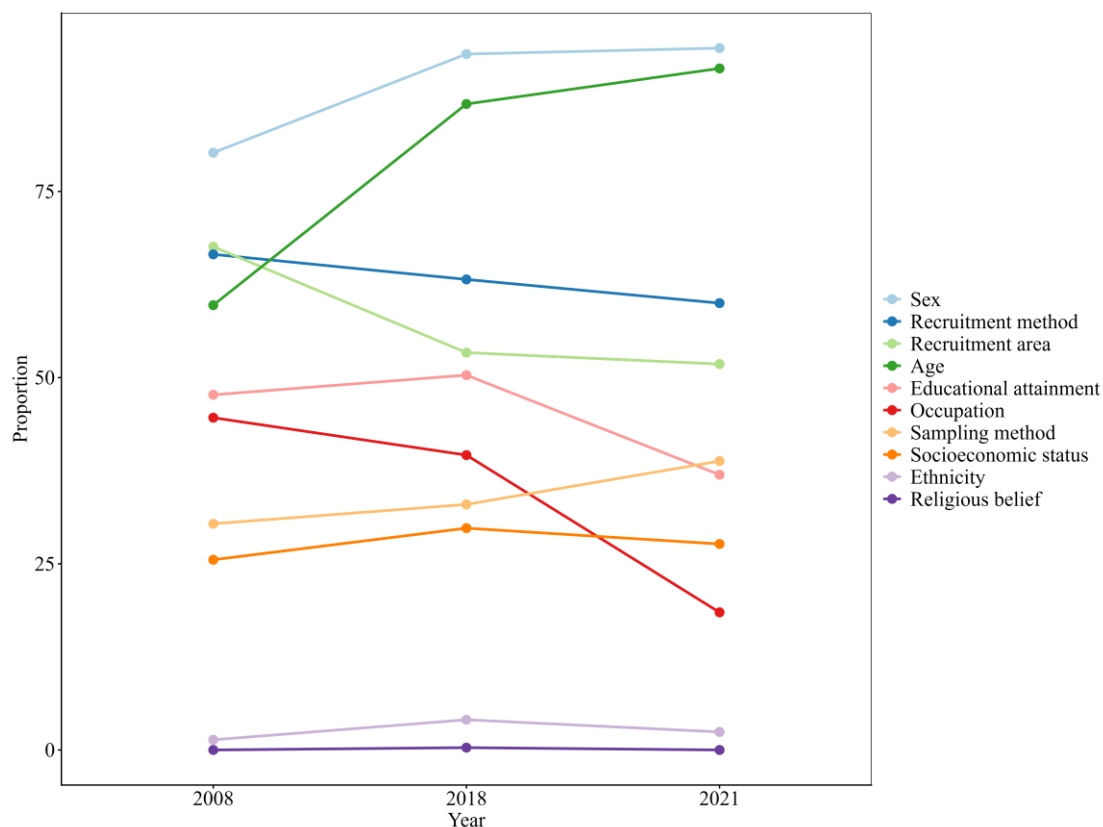


Fig S3. Proportion of participants with demographic information reported in five Chinese journals across different periods.

Among the 1,000 articles, 622 targeted special populations and 378 targeted the general population (only 24 articles explicitly stated this, and 354 were inferred in our coding process). The reporting of demographic information for both general and specific populations is presented in Fig S4. For those aiming to generalize to specific populations, 117 articles targeted children and teenagers, 107 articles targeted college students and graduate students, 62 articles targeted primary or secondary school students, 37 articles targeted infants and young children, and 300 articles targeted other special populations (Fig S5).

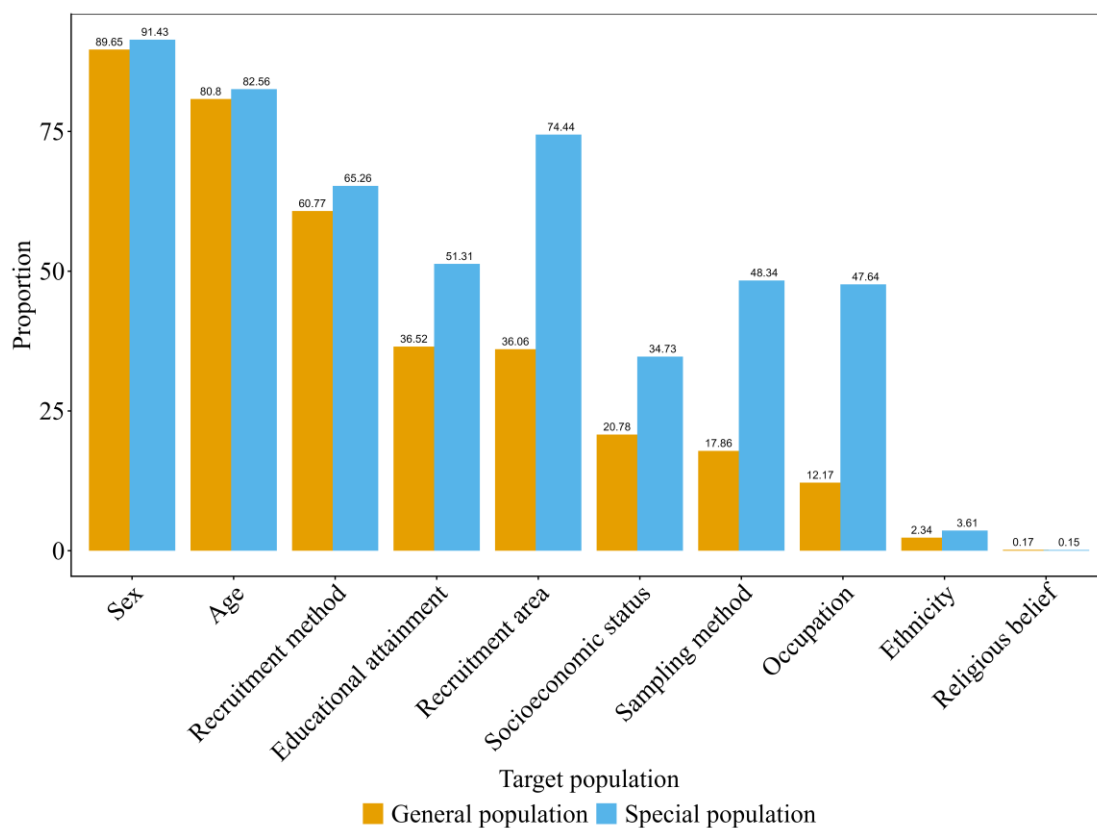


Fig S4. Proportion of participants with demographic information reported across five Chinese journals by the target population

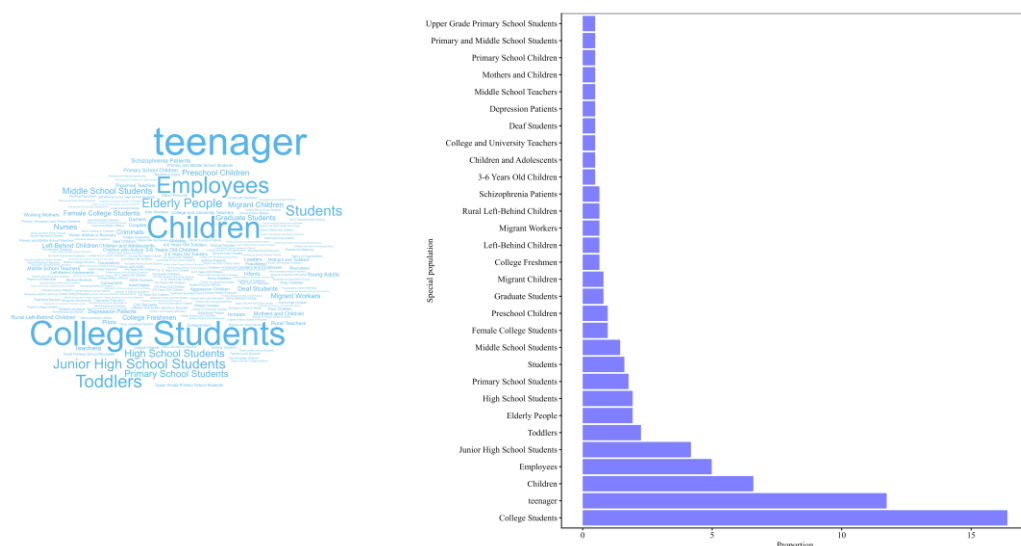


Fig S5. The word cloud of the special population (left figure) and the proportion of the special population among all special population (right figure). In the word cloud, identical colors denote identical frequencies, and larger font sizes correspond to higher frequencies, whereas the bar chart displays only population groups with a frequency greater than 1.

S6. Supplementary results: Age distribution of the three hypotheses

The results 7-bins approach for age distribution are presented below.

For our first research hypothesis, in the general population, we found that the large-scale international collaborative projects differ from the Chinese papers, with the extreme evidence from the Bayesian multinomial test ($\ln(BF_{a0}) = 14.45$). For our second research hypothesis, we also found that extreme evidence in favor of the age distribution of the pooled data of Chinese participants is different from that of the 7th Census data ($\ln(BF_{a0}) = 63.21$). For our third research hypothesis, the pairwise Bayesian multinomial test revealed moderate evidence that samples from 54 countries and regions are the same as Chinese psychological samples (see Fig S6A and Fig S6B).

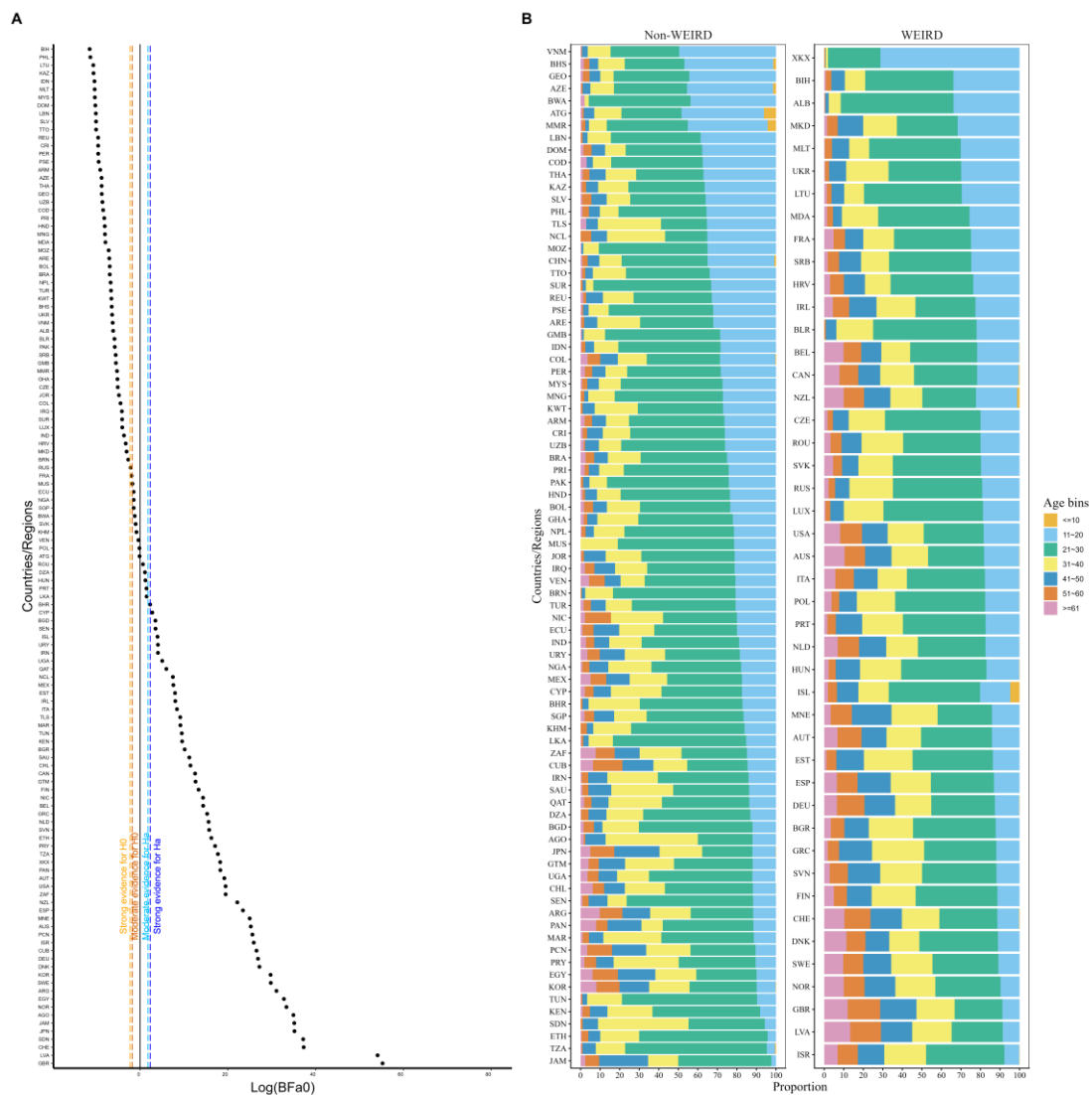


Fig S6A and Fig S6B. The age distribution from psychological participants, with 10-year as intervals. (A) Pairwise comparisons of age distribution between Chinese psychological participants and available data from other countries and regions; (B) Age proportion of all data. Country code is ISO3.

The visualization of the age distribution at developmental stages and the pairwise Bayesian multinomial test are shown in Fig S7.

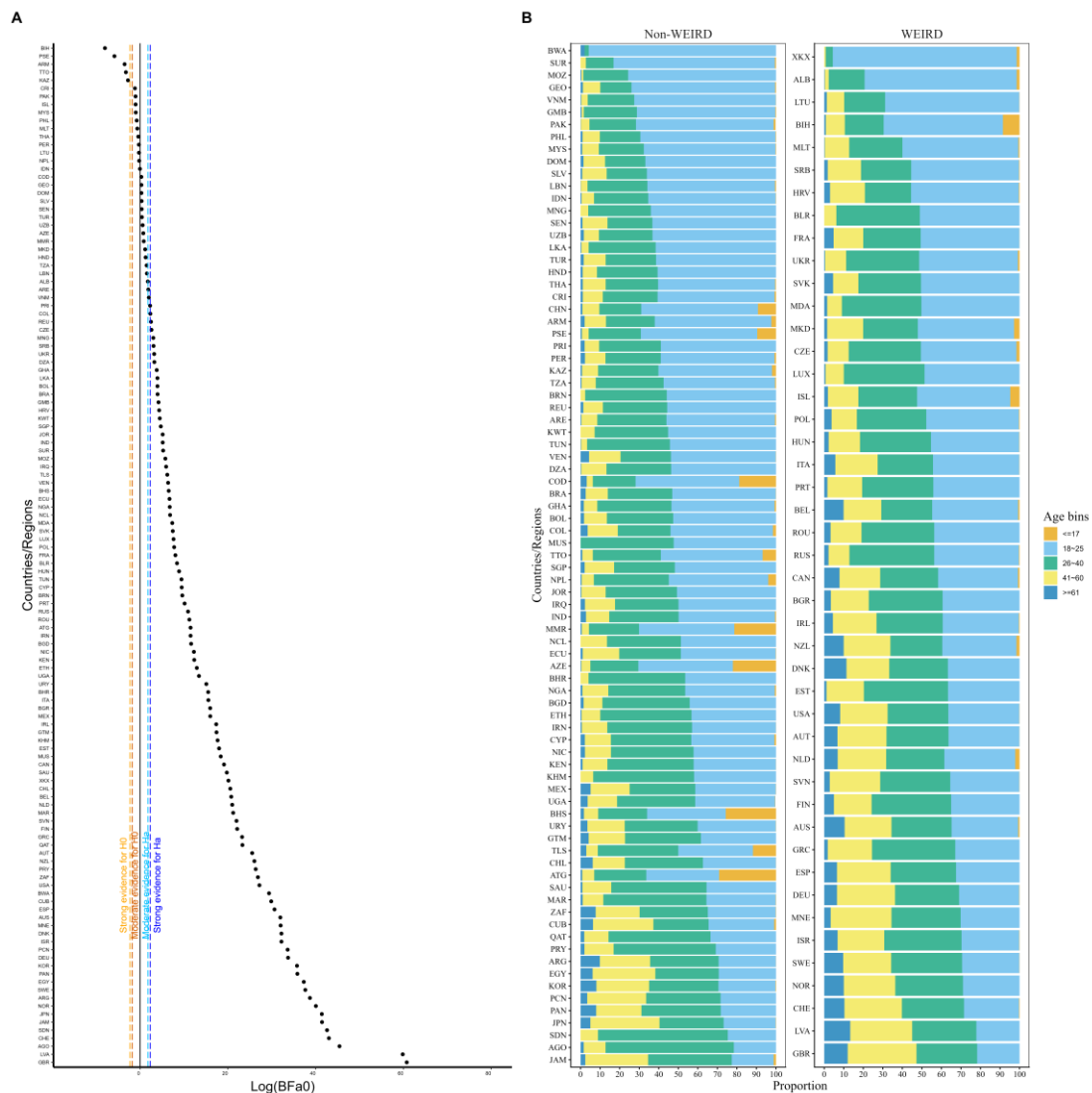


Fig S7. The age distribution in psychological participants. (A) Pairwise comparisons of age distribution between the Chinese psychological sample and available data from other countries and regions; (B) Age ratio of all data. Country code is ISO3.

S7. Supplementary results: Comparison of each data source with the Census data and CFPS' data

For the sex ratio distribution, we compared the Chinese papers and large-scale international collaborative projects with the Census data and data of CFPS, respectively (Fig S8). Our results suggested that there is moderate evidence supporting H_0 across all comparisons (Journal vs. Census6, $\ln(BF_{a0}) = -1.44$; BTS Chinese vs. Census6, $\ln(BF_{a0}) = -2.06$; Journal vs. Census7, $\ln(BF_{a0}) = -1.44$; BTS Chinese vs. Census7, $\ln(BF_{a0}) = -2.07$; Journal vs. CFPS2018, $\ln(BF_{a0}) = -1.68$; BTS vs. CFPS2018, $\ln(BF_{a0}) = -2.08$).

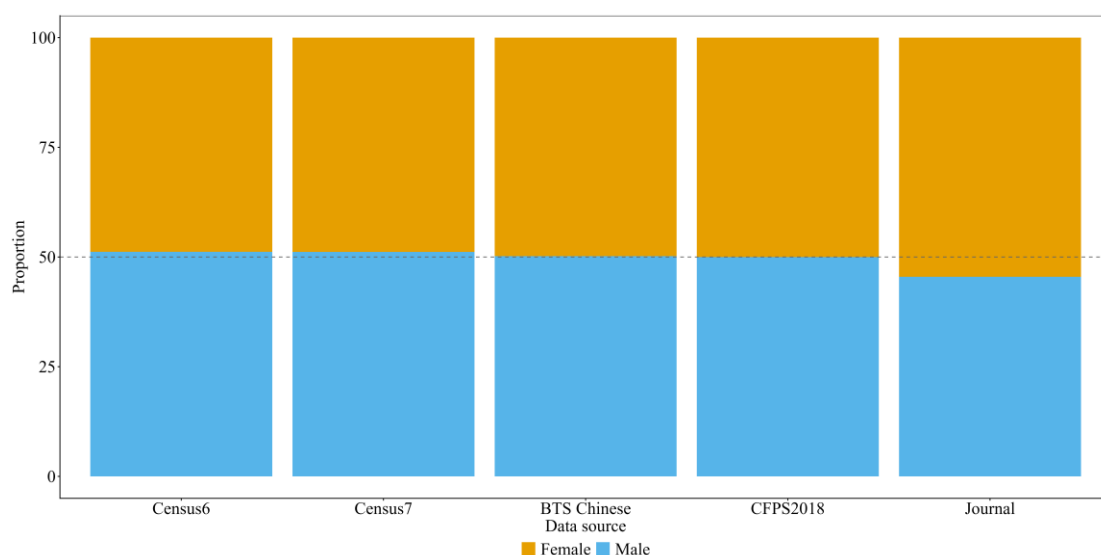


Fig S8. Large-scale international collaborative projects, Journal papers, Census 6th and Census 7th data on the sex ratio distribution. BTS Chinese: the Chinese participants from large-scale international collaborative projects; Journal: the Chinese participants from Chinese papers; CFPS2018: China Family Panel Studies 2018.

For the age distribution, we also compared the two data sources with China's census data, respectively (see Fig S9 and Fig S10). It was found that the age group of 20~24 years old accounts for the highest proportion in both data sources, and this result is consistent with that in the main text. Meanwhile, we also compared the two data sources with age classification approaches (5-bins and 7-bins) of the census data respectively, and all results indicated that there is extreme evidence to prove that participants in psychological research differ from those in the Census data (5-bins approach for age: Journal vs. Census6, $\ln(BF_{a0}) = 65.73$; BTS

Chinese vs. Census6, $\ln(BF_{a0}) = 43.82$; Journal vs. Census7, $\ln(BF_{a0}) = 99.42$; BTS Chinese vs. Census7, $\ln(BF_{a0}) = 67.62$. 7-bins approach for age: Journal vs. Census6, $\ln(BF_{a0}) = 52.92$; BTS Chinese vs. Census6, $\ln(BF_{a0}) = 25.50$; Journal vs. Census7, $\ln(BF_{a0}) = 75.03$; BTS Chinese vs. Census7, $\ln(BF_{a0}) = 41.65$.

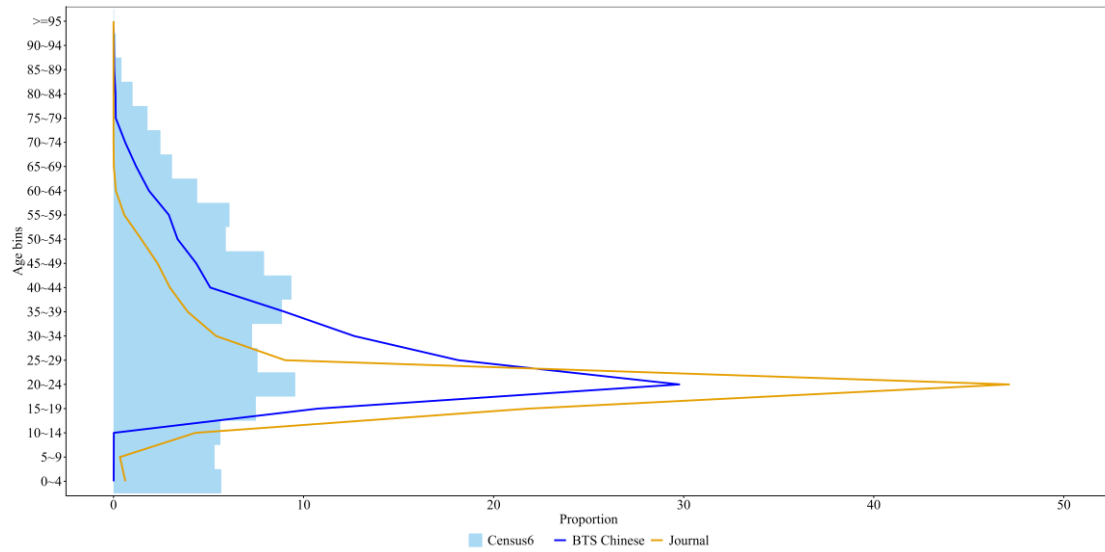


Fig S9. Large-scale international collaborative projects, Journal papers, and the 6th Census data on the age distribution.

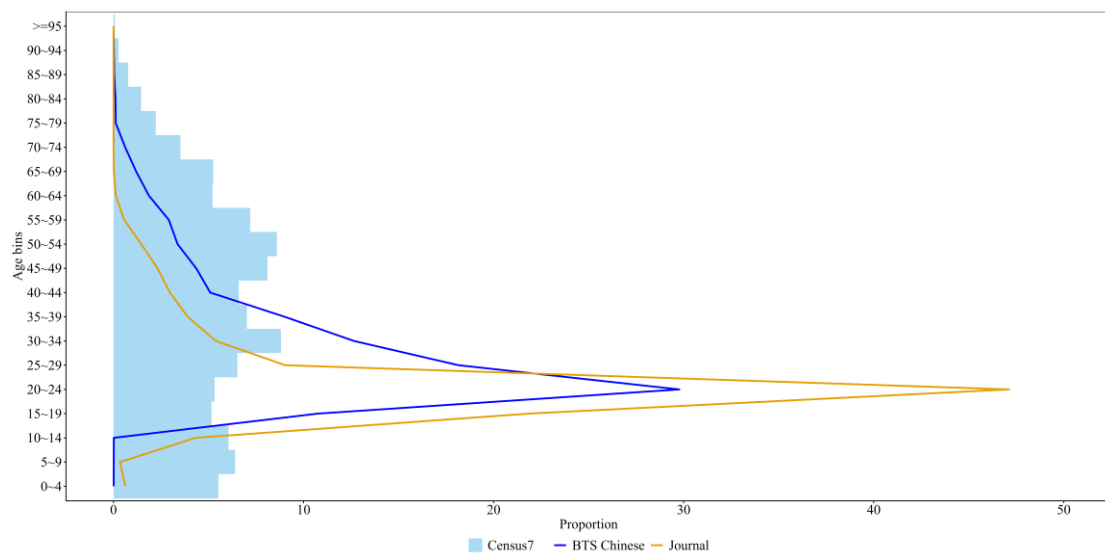


Fig S10. Large-scale international collaborative projects, Journal papers, and the 7th Census data on the age distribution.

We also compared the educational attainment data from large-scale international collaborative projects and Chinese papers separately against the Census data and data of CFPS 2018, using two cutoffs (college-level and high-school level). For the college-level cutoff, extreme evidence showed differences between the large-scale international collaborative projects sample and Census data (Census6: $\ln(BF_{a0}) = 148.22$; Census7: $\ln(BF_{a0}) = 101.95$), and between the Chinese journal sample and Census data (Census6: $\ln(BF_{a0}) = 160.61$; Census7: $\ln(BF_{a0}) = 112.33$). For the high school-level cutoff, similarly extreme evidence emerged for both comparisons (BTS Chinese vs. Census6: $\ln(BF_{a0}) = 113.65$; BTS Chinese vs. Census7: $\ln(BF_{a0}) = 84.81$; Journal vs. Census6: $\ln(BF_{a0}) = 95.76$; Journal vs. Census7: $\ln(BF_{a0}) = 68.95$).

S8. Supplementary results: Regional comparison of Chinese psychological participants using 7th Census data and CFPS' data

To compare the regional distribution of Chinese psychological participants targeted at the general population, 7th Census data, and the data of CFPS 2018, we mapped the regional distribution, as illustrated in Fig S11.

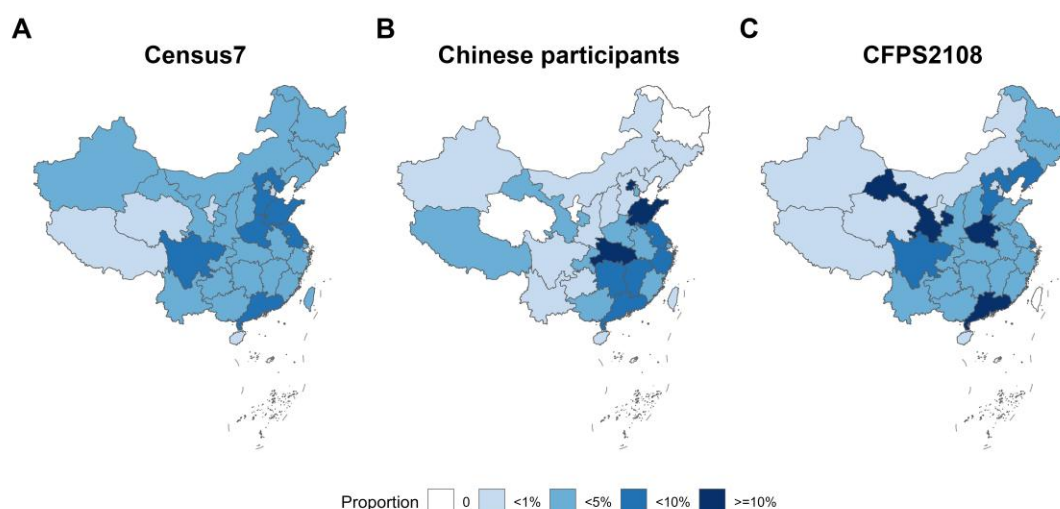


Fig S11. The regional distribution of Chinese papers, 7th Census data and the data of CFPS 2018.

S9. Supplementary results: The “default participants” reported in the abstract

Psychological studies often rely on college students, and researchers may treat them as the “default participants” and choose not to mention them in the abstract. Therefore, we examined whether authors mentioned "college students" in the abstracts when the studies included college student samples. The analysis was conducted in JASP 0.95.4 only for articles that focus on the general population (JASP Team, 2025). Among these articles, we found that there were more articles not mentioning "college students" in their abstract ($n = 225$) than those mentioning ($n = 80$), $\ln(BF_{a0}) = 33.13$.

We further examined differences in the reporting of demographic information across different sample types (Table S3). We found that when samples are all college students, there are more studies did not report the sample information in their abstract than those reported ($\ln(BF_{a0}) = 17.29$). Conversely, for those studies are only from samples outside colleges, more studies

reported sample information in the abstract than those that did not, $\ln(BF_{a0}) = 3.93$. For those studies that used both college students and those outside college, we did not find evidence for a difference between them, $\ln(BF_{a0}) = -0.98$. These results suggested that researchers may treat college students as the default participants when they are interested in generalizing their conclusions to the general population.

Table S3. Different study samples and their sample mentions in the abstract within the general population.

Study sample	Samples not mentioned	Samples mentioned	Total
Only college students	197	91	288
Only sample outside colleges	11	33	44
College students & other samples	7	10	17

S10. Supplementary results: Geographic representation of participants

To investigate whether different study types affect the distribution of participant recruitment regions, we analyzed the differences between experimental studies and questionnaire studies across two regional classifications. All results indicated very strong/extreme evidence of disparities between the data from Chinese psychological studies and both the Census data and the data of CFPS 2018 (Table S4). See the spatial distribution of participant source regions across different study types in Fig S12.

Table S4. Different study types and their differences in participants' recruitment regions

Comparative categories	$\ln(BF_{a0})$ in four regions	$\ln(BF_{a0})$ in seven regions
All studies vs Census6	10.21	4.69
Experiment studies vs Census6	20.66	9.15
Questionnaire studies vs Census6	8.58	7.25
All studies vs Census7	8.71	4.13
Experiment studies vs Census7	17.91	7.72
Questionnaire studies vs Census7	7.77	7.20
All studies vs CFPS2018	19.79	22.23
Experiment studies vs CFPS2018	31.13	25.94
Questionnaire studies vs CFPS2018	17.76	25.36

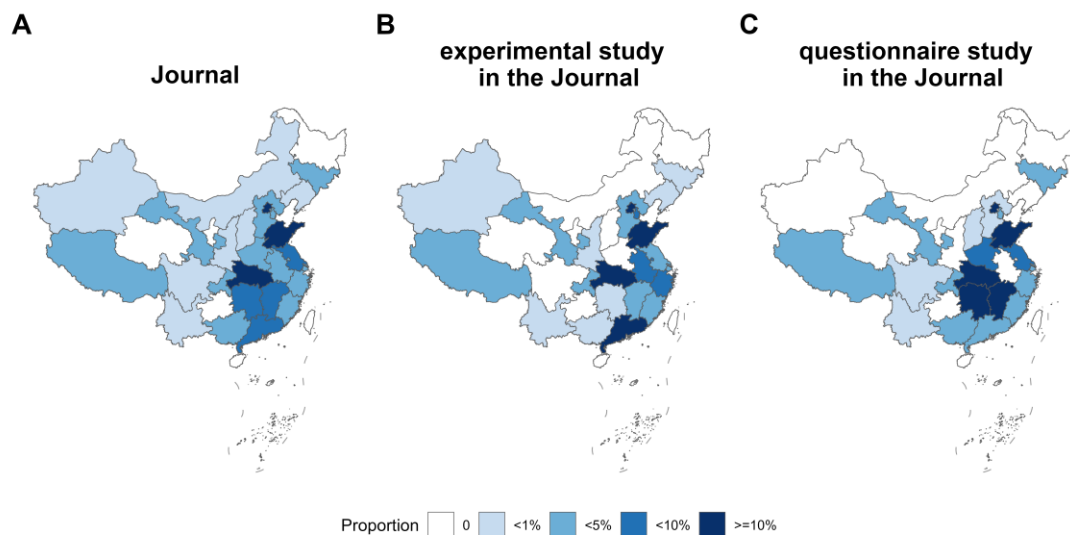


Fig S12. The participants' recruitment region distribution in different study types

S11. Supplementary results: The spatial correlation between the ranking of psychology departments and sample size.

We examined the relationship between the 2021 Shanghai Ranking (a leading higher education evaluation agency, website: <https://www.shanghairanking.cn/>) of Departments/Schools of Psychology and/or Applied Psychology and the provincial sample sizes reported in Chinese journal articles. The 2021 rankings were selected for two reasons. First, Shanghai Ranking launched its University Major Rankings (mainland China) that year, which remains the most comprehensive assessment of university majors in mainland China and includes both Psychology and Applied Psychology. Second, the 2021 data are temporally closest to the journal article sample used in our study.

To simplify the question, we used the rank of the highest-ranked Psychology/Applied Psychology Department in each province (Psychology: $n = 21$; Applied Psychology: $n = 29$). Note that Hong Kong, Macao, and Taiwan provinces were excluded from the analyses because the ranking focused on institutes in mainland China). For example, Nanjing Normal University and Jiangsu Normal University are both located in Jiangsu province, but the former was ranked 6th, and the latter ranked 42nd, we used Nanjing Normal University's rank for Jiangsu province. We also ranked the provinces by their sample size in 1000 psychological articles. We then calculated the ranks of these provinces using Spearman's correlation.

The Bayes factors for Spearman's correlation analysis revealed that there was strong evidence of an association between the Applied Psychology ranking and provincial sample sizes reported in the journals ($\ln(BF_{a0}) = 2.99$). Similarly, anecdotal evidence of an association between the Psychology ranking and provincial sample sizes ($\ln(BF_{a0}) = 0.75$). This result was not surprising, as psychological researchers usually do not recruit participants from spatially remote areas from their home institutes.

S12. Supplementary results: Comparison of themes between Chinese papers and large-scale international collaborative projects

We analyzed keywords from 1,000 Chinese papers and 27 large-scale international collaborative projects. As the keywords for some international projects were not available, we could only retrieve keywords from 14 projects. As shown in Fig S13, 69 keywords were extracted from these 14 projects, with 7 keywords appearing more than twice. The word "Culture" occurs 4 times and is ranked highest. For Chinese journal papers, 84 keywords were selected¹, where "college students" (61 times) and "teenager" (47 times) emerged as high-frequency terms. It is worth noting that only one keyword, positive emotions, is present in both datasets. Given the considerable disparity in sample sizes between the two data sources, this study only presents descriptive statistics and visualization results, without further statistical inference.

Fig S13. The word cloud of international collaborative projects (left figure) and Chinese journal papers (right figure). Keywords in the right figure were translated from Chinese into English.

¹ From the 65th to the 84th keywords, the frequency of each is 5.

Reference

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